

STAT 380 – Homework 1 (Data Visualization)

HW1 is due by 11:59 PM on **Thursday**, January 23, 2025.

Learning Objectives

This assignment address aspects of the following learning objectives.

1. Students will be able to load datasets from R packages ~~and external sources~~ into the R environment.
2. Given one or more variables, students will be able to identify appropriate types of visualizations for exploring the distribution of the variable or the relationship between the variables visually.
3. Given a research question, students will be able to create and customize an appropriate visualization using statistical software.
4. Given a visualization ~~and summary statistics~~, students will be able to describe the insights gained from the information.

The crossed-out parts of 1. and 4. are not covered on this assignment.

Goal

This goal of this activity is to practice creating data visualizations using **ggplot**. Further, this activity will help you to practice searching for and implementing techniques to customize ggplot visualizations. You will not be able to memorize all R code. The trick is learning some basics ideas and terminology well enough that you can search the internet for solutions.

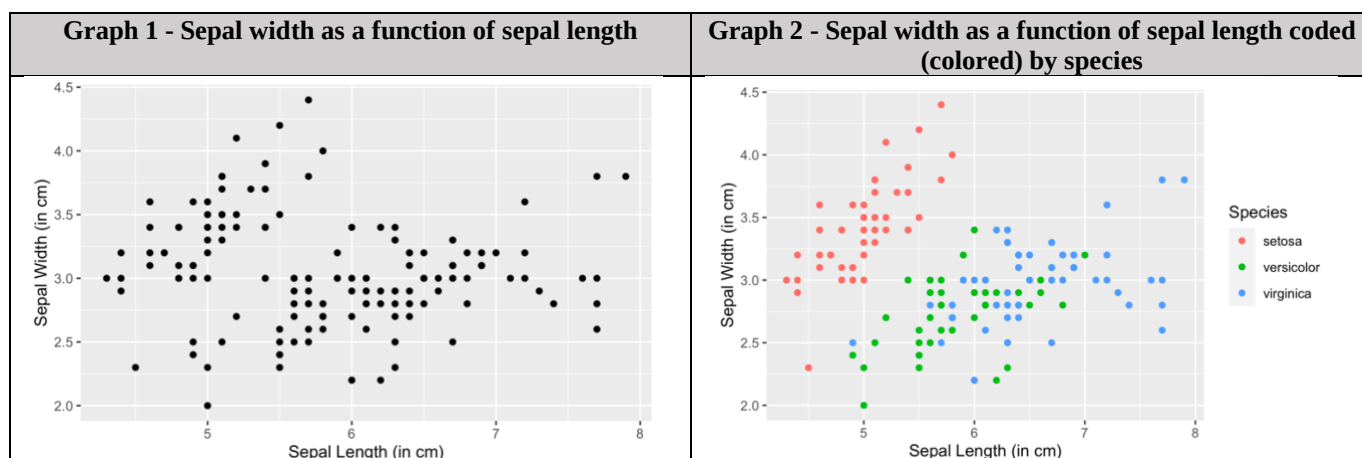
Create a new R Notebook by going to FILE >> NEW FILE >> R Markdown. Modify the YAML header in accordance the details in Example 1 of Lecture 1. In the YAML header, title the document “STAT380 HW1” and be sure to include the date and your name. Following the YAML header, delete the default text, add a section titled Front Matter (using a level 2 heading and add libraries as needed – be sure the libraries show up in your html document), and save the file as LastnameFirstinitial_380HW1.Rmd.

This activity uses a dataset that is part of Base R (collection of packages that are automatically loaded every time you begin a new R session). The dataset is called `iris`. You can create of a copy of iris in your environment using the command `data(“iris”)`. NOTE: If you copy from the Word document into R, R will not recognize the quotation marks as proper symbols. Instead, type the command yourself in R.

The iris dataset contains information (measurements) for 3 species of iris flowers. A snapshot of the dataset is shown below:

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa

A few basic plots are shown below:

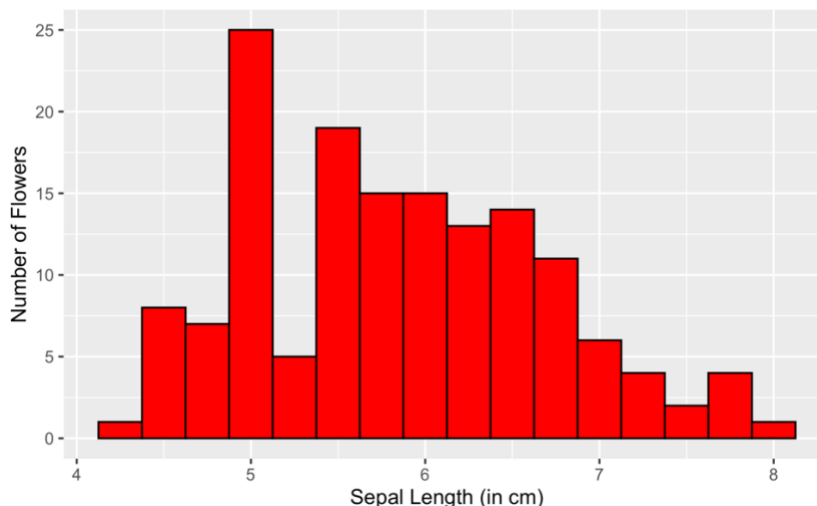


Problems

Some of the problems that follow will closely follow ideas covered in the notes. Other problems will either require prerequisite knowledge from STAT 184 or searching the internet to find solutions. In addition to reviewing the notes, one goal of this exercise is to practice your ability to find solutions online. This will be one of the most valuable skills you develop!!!

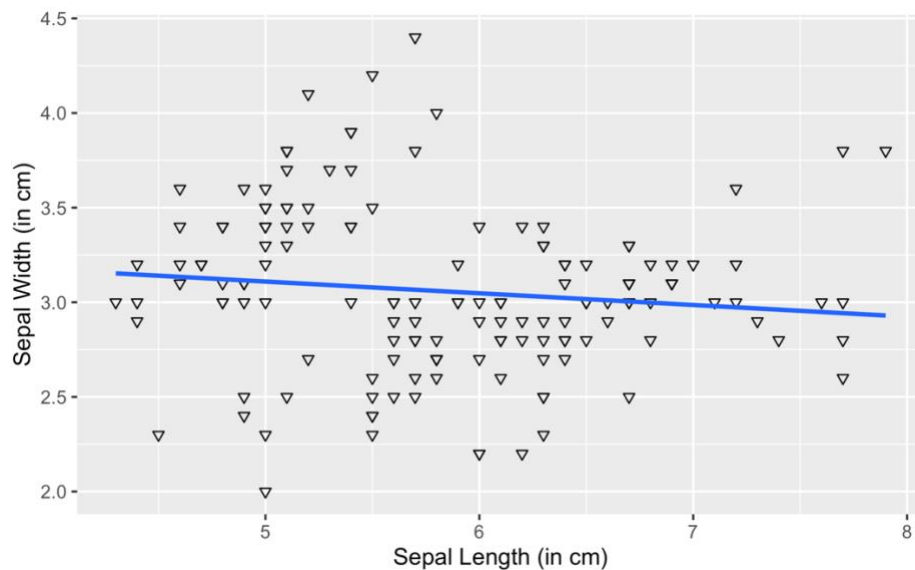
For each problem, add a level two header with the problem number (e.g., Problem 1, Problem 2, Problem 3, etc.), insert R chunks as needed, and include any answers to questions as text outside of R chunks.

Problem 1 Suppose we want to explore the distribution of the sepal length variable. Recreate the histogram shown below. **Then**, comment on the shape of the distribution. Some details: Use a binwidth of 0.25. The bars are red with a black outline. NOTE: When asked to recreate a plot, your plot should match exactly. In this problem, that means matching axis titles, gray background, red bars, black outlines, bin width of 0.25, done using `ggplot`, etc.

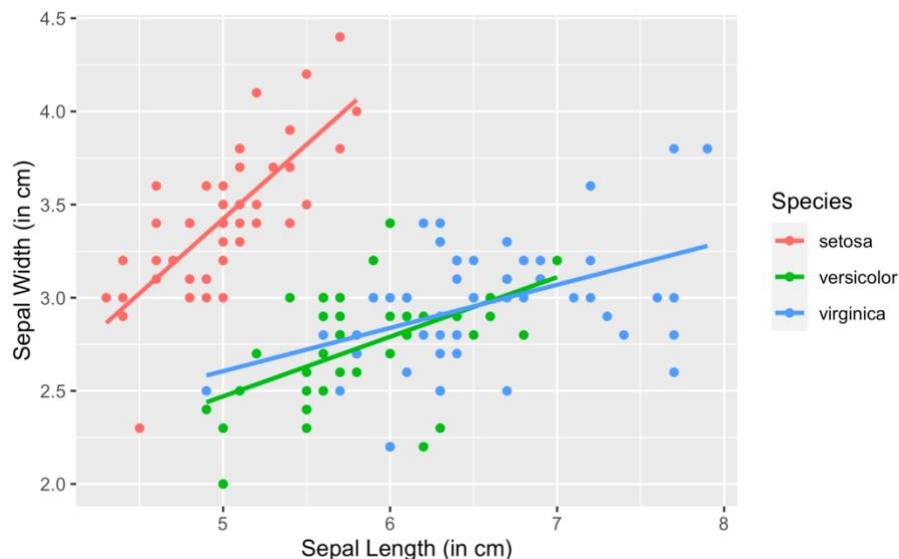


Problem 2 Create an appropriate plot that shows the distribution of Species. **Then**, comment on what you have learned about the Species variable from the plot.

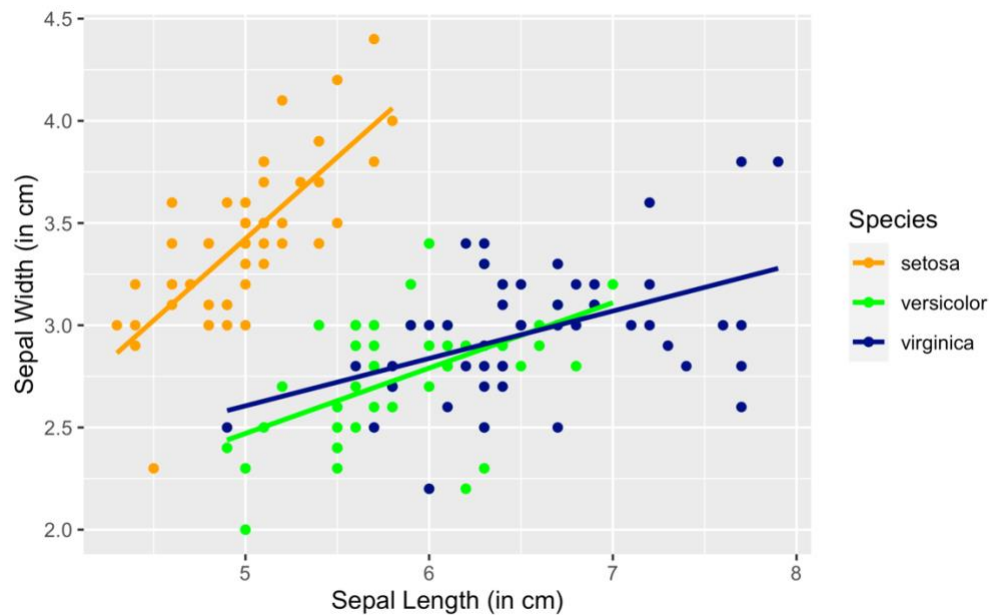
Problem 3 Research Question: Is there a linear relationship between sepal width and sepal length? To answer this, recreate Graph 1 (scatterplot of sepal width (y) as a function of sepal length (x) which is shown on the top of Page 2) but make two changes: i) change the plotting symbols to be an open triangle point down and ii) add a regression trend line as shown below. ***Then***, comment on what you have learned about the relationship between sepal length and sepal width based on the plot. HINT: You are unlikely to know which shape value produces the intended plotting symbols. I did a Google search for “r shape values ggplot”. The second link <http://www.sthda.com/english/wiki/ggplot2-point-shapes> was useful and included examples. In future questions, you will write your own search query.



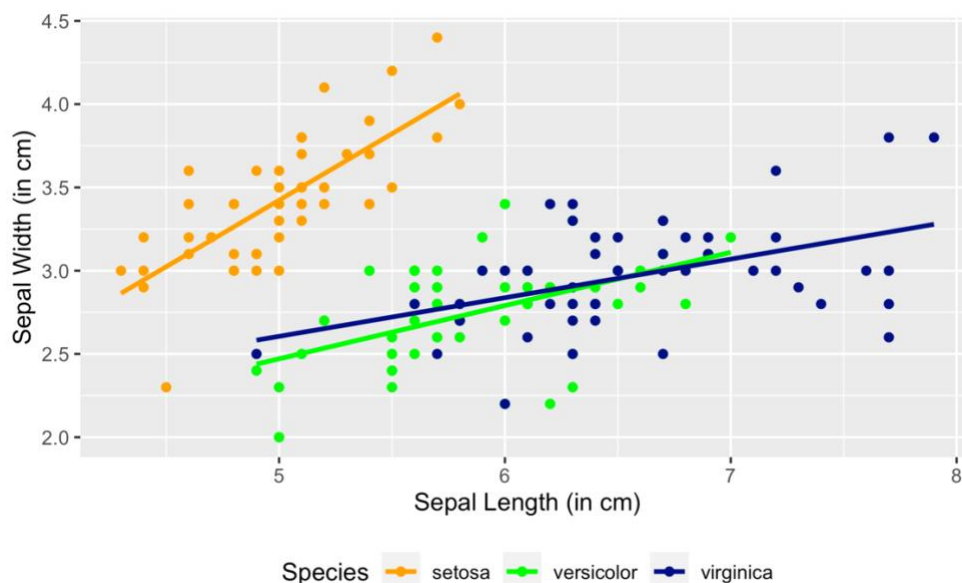
Problem 4 The relationship between sepal width and sepal length may be affected by other variables that we have not considered. Revised research question: Does the relationship between sepal width and sepal length depend on the species? For instance, the species of the flower may impact the relationship between sepal width and sepal length. To answer this question, recreate Graph 2 (scatterplot showing sepal width as a function of sepal length coded (colored) by species as shown on Page 2) and add a regression trend line for each species as shown below. ***Then***, comment on what you have learned about the relationship between sepal length and sepal width based on the plot.



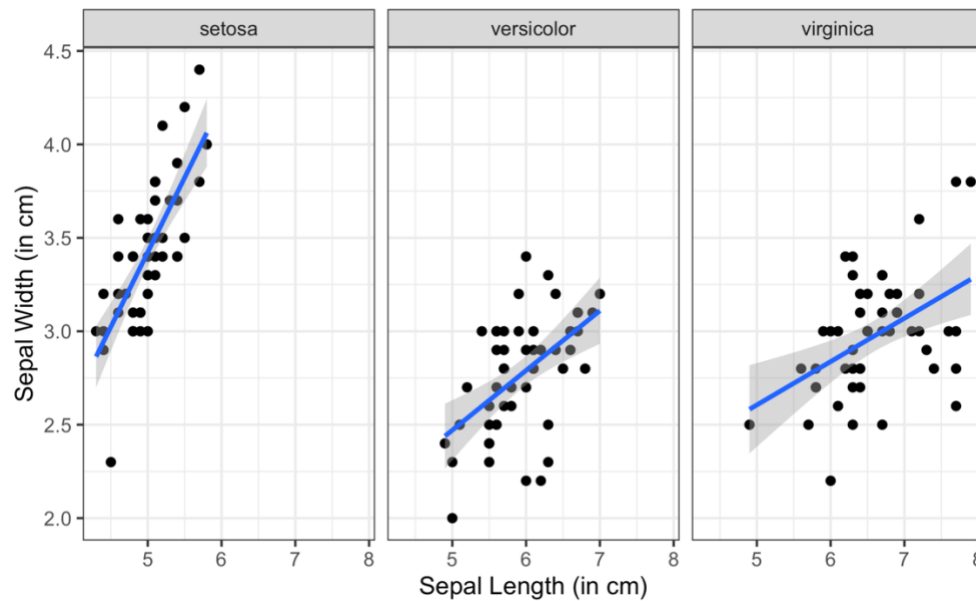
Problem 5 Recreate the plot from the previous question but change the color scheme for the different species from the default colors to orange, green, and navy as shown below. **In addition to producing the plot,** include the search query you used to change the color scheme **and** a link to a website that showed you how to do this. (Even if you knew how to do this, humor me, do a search, and provide a link.) NOTE: If you used generative AI rather than a search engine, provide the prompt used and the name of the tool (ChatGPT, Co-pilot, etc.). In such cases, a link is not necessary.



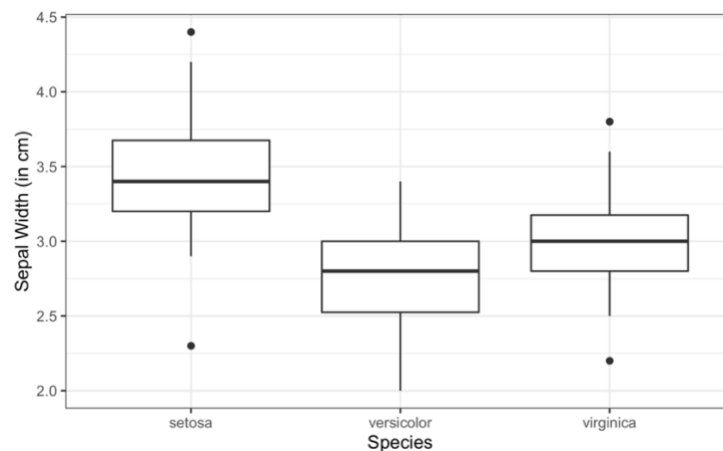
Problem 6 For the plot created in the previous question, change the legend location to the bottom. **In addition to producing the plot,** include the web search you used to change the legend location **and** a link to a website that showed you how to do this. (Even if you knew how to do this, humor me, do a search, and provide a link.) If you used generative AI rather than a search engine, provide the prompt used and the name of the tool (ChatGPT, Co-pilot, etc.).



Problem 7 Create a new plot that shows sepal width as a function of sepal length, facets based on species, adds a regression trend line (use linear, not loess; include error boundaries), and removes the gray plot background as shown below. **In addition to producing the plot**, include the web search you used to remove the gray background **and** a link to a website that showed you how to do this. (Even if you knew how to do this, humor me, do a search, and provide a link.) If you used generative AI rather than a search engine, provide the prompt used and the name of the tool (ChatGPT, Co-pilot, etc.).



Problem 8 Create side-by-side boxplots showing the sepal width broken down by species. Remove the gray background.



At the conclusion of the Homework 1, you should upload the following files to Canvas:

1. your .html file named LastnameFirstinitial_380HW1.html (or your .nb.html if using R Notebook)
2. your .Rmd file named LastnameFirstinitial_380HW1.Rmd

NOTE: If you could not get your .Rmd file to Knit properly, you will receive some partial credit, but there will be a penalty if the .html file is missing, incomplete, etc. **Be sure to proofread the .html file before uploading. You are responsible for making sure you have uploaded the correct file that has all of your answers.**